MASTER SYLLABUS

Physics Self-Learning Plan

Summer 2026 – Spring 2027

Owner:	Parker
Summer budget:	10 hrs/week
Fall–Spring budget:	20 hrs/week
Total topics:	12 (all on review list)
Total weeks scheduled:	53

Twelve physics topics consolidated from the 2026–27 self-study syllabi and the grad-review plans, restructured into six study periods. Every weekly entry lists reading + a specific problem set with explicit hour budgets that sum to the period's per-subject weekly slice.

Compiled May 15, 2026

Periods at a Glance

Period	Dates	Wks	Hrs/wk	Subjects	Per-subj hrs/wk
Summer #1	May 15 – Jul 15, 2026	9	10	Nuc #2 · Stat Mech · Exp HEP Pt 1	≈3.3 each
Summer #2	Jul 22 – Aug 18, 2026	4	10	Nuc #1 · Exp HEP Pt 2	≈5 each
Fall #1	Sep – Oct 2026	9	20	Quantum · RHI & Jets · E&M	≈6.7 each
Fall #2	Nov – Dec 2026	9	20	QFT #1 · Classical Pt 1	QFT 12 / CM 8
Spring #1	Jan – Feb 2027	9	20	QFT #2 · Classical Pt 2	QFT 12 / CM 8
Spring #2	Mar – May 2027	13	20	Astro & Nuc Astro · Optics	Astro 12 / Opt 8

Per-Subject Weekly Time Budgets (reading vs. problems)

Subject	Total / wk	Reading	Problems
Nuc #2, Stat Mech, Exp HEP Pt 1	3.3 hrs	2.0 hrs	1.3 hrs
Nuc #1, Exp HEP Pt 2	5.0 hrs	3.0 hrs	2.0 hrs
Quantum, RHI & Jets, E&M	6.7 hrs	4.0 hrs	2.7 hrs
QFT #1, QFT #2	12 hrs	7.0 hrs	5.0 hrs
Classical Pt 1, Pt 2	8 hrs	5.0 hrs	3.0 hrs
Astro & Nuc Astro	12 hrs	7.0 hrs	5.0 hrs
Optics	8 hrs	5.0 hrs	3.0 hrs

Every week below lists the same split. If the week comes in short, drop reading before problems — problem-solving is the retrieval practice that builds recall.

Topic Coverage

#	Topic (review-list order)	Period	Source plan
1	Nuc #1 – Decay, reactions, fission, detectors	Summer #2	May–Jul Nuc, Wks 1–4
2	Nuc #2 – Particle physics, EW, Higgs, neutrinos	Summer #1	May–Jul Nuc, Wks 5–10
3	Stat Mech – Ensembles to Brownian motion	Summer #1	Grad Review, SM track
4	Quantum – Merzbacher review	Fall #1	Grad Review, QM track
5	E&M – Zangwill review	Fall #1	Grad Review, E&M track
6	Classical – Lemos (Lagrangian → Hamiltonian)	Fall #2 + Spring #1	Grad Review, CM track
7	QFT #1 – Canonical → tree-level QED	Fall #2	Class 1 (QFT I)
8	QFT #2 – Path integrals, renormalization, SM	Spring #1	Class 2 (QFT II)
9	RHI & Jets – QGP → HEP jets	Fall #1	Class 3 (First-Exposure)
10	Exp HEP – Detectors, accelerators, landscape	Summer #1 + #2	Class 5
11	Astro & Nuc Astro – Stars → cosmology	Spring #2	Class 4
12	Optics – Classical → quantum optics	Spring #2	Class 6

Cuts vs. Original Syllabi

To fit the new hour budgets, the following were trimmed:

- **QFT I & II:** P&S core kept; cut Weinberg / Srednicki side-reads and most supplementary problem sets. One capstone per QFT class.
- **RHI & Jets:** use First-Exposure edition; ML-for-jets (Wk 9 original) and saturation/EIC (Wk 12 original) reduced to recommended reading.
- **Astro:** kept Foundations primer + stellar/compact/cosmology weeks; cosmic-rays cut to a single combined week.
- **Exp HEP:** split across two summers. Pt 1 = detectors + statistics + landscape. Pt 2 = accelerators + TDR deep-dives + capstone.
- **Optics:** cut aberrations week; merged interference into one combined block.
- **Classical:** dropped Hamilton–Jacobi (Lemos Ch. 9) and classical field theory (Ch. 11) — covered by QFT.
- **Nuc #1:** dropped Loveland Ch. 14 (reactors/accelerators) — overlaps Exp HEP.

> **Workflow per week per subject:** skim → AI LaTeX notes → reread with margin answers → tackle the listed problems. The hour split below assumes a ∼60/40 reading-to-problem ratio. If you finish problems early, add one more from the same chapter; if you cannot finish one in 1.5× its expected time, stop and write what blocked you.

Summer #1

May 15 – Jul 15, 2026 · 9 weeks · 10 hrs/wk · ∼3.3 hrs/subj/wk

> Three subjects in parallel is tight. Each week pick one subject as the "deep" subject (target ∼5 hrs); keep the other two on maintenance pace (∼2.5 hrs). Rotate the deep slot. Weeks 8–9 are buffer + capstone.

Nuc #2 — Particle Physics & Electroweak / Neutrino

Anchor: Mann Chs. 12, 23, 25; Thomson Ch. 13. Target: own the gauge → EW → Higgs → ν-mass arc.

Wk 1. Particle-physics foundations
Reading Mann Chs. 3, 4, 5, 6, 9, 10 (∼96 pp) — 2 hrs
Problems Mann Ch. 3 Q1; Ch. 4 Q2; Ch. 5 Q2; Ch. 6 Q3; Ch. 9 Q2; Ch. 10 Q3 (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 2. QFT propagators & S-matrix
Reading LB Chs. 7, 12, 13, 16, 17, 18, 19, 20 (∼66 pp) — 2 hrs
Problems LB Ex. 17.1, 18.1, 19.1, 19.2, 20.1, plus 7.1 bonus (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 3. Gauge invariance & QED
Reading Mann Chs. 11, 12, 13, 14 skim (∼60 pp) — 2 hrs
Problems Mann 11 Q2; 12 Q1; 13 Q3; 14 Q1 (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 4. Weak interactions
Reading Mann Chs. 19, 20, 21 + LB Ch. 26 (SSB preview) — 2 hrs
Problems Mann 19 Q3; 20 Q2; 21 Q2; LB Ex. 26.1 (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 5. ⋆ Higgs mechanism & EW breaking
Reading Mann Chs. 22, 23 + LB 46, 47 + Thomson 17 (skim) — 2 hrs
Problems Mann 22 Q2; 23 Q1; 23 Q3; LB 46.1; LB 47.1 (1.3 hrs) — this is the chapter to memorize
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 6. ⋆ Neutrino oscillations
Reading Mann 24 + 25 + Thomson 13 + Thomson §17.8 — 2 hrs
Problems Thomson 13.1, 13.3, 13.5; Mann 24 Q1; Mann 25 Q1 (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 7. Synthesis — write up Higgs derivation
Reading Re-read Mann 23 §23.2 + LB 47 + draft capstone outline — 2 hrs
Problems Re-derive W, Z masses from $v \approx 246$ GeV without notes; derive $P(\nu_e \to \nu_\mu)$ from scratch (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 8. Capstone draft
Reading Skim recent ATLAS / CMS Higgs paper (e.g. H→ $\gamma\gamma$ Run 2) — 2 hrs
Problems Draft 1-page capstone: *gauge invariance → EW unification → Higgs → ν-mass* (1.3 hrs)
Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 9. Capstone final + slip buffer
Reading Final polish, fact-check (1.5 hrs)
Problems Tackle deferred problems from weakest week (1.8 hrs)
Time Reading 1.5 + Problems 1.8 = **3.3 hrs**

Milestones: end of Wk 5 derive m_W, m_Z from VEV without notes; end of Wk 6 derive 2-flavor oscillation P.

Stat Mech — Ensembles to Brownian Motion

Anchor: all three ensembles + BE/FD + phase transitions + Brownian motion. Schroeder for intuition, Pathria for formal machinery.

Wk 1. Energy & 2nd law

Reading Schroeder Chs. 1–2 (∼50 pp) — 2 hrs

Problems Schroeder 2.18 (Einstein solid multiplicity); 2.22 (entropy increase) (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 2. ⋆ Microcanonical ($\Omega \to S \to T$)

Reading Schroeder Ch. 3 + Pathria §2.1–2.5 — 2 hrs

Problems (a) two Einstein solids in thermal contact, derive T from $\partial S/\partial U$; (b) derive Sackur–Tetrode for ideal gas from $\Omega(E)$; (c) Pathria 2.x: Ω for SHO ensemble (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 3. ⋆ Canonical ensemble

Reading Pathria Ch. 3 §3.1–3.7 + Schroeder Ch. 6 §6.1–6.3 — 2 hrs

Problems (a) compute Z and all thermo for 2-level paramagnet; (b) classical rotor (translational/rotational/vibrational Z factorisation); (c) verify $\langle(\Delta E)^2\rangle = k_B T^2 C_v$ from Z (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 4. ⋆ Grand canonical

Reading Pathria Ch. 4 §4.1–4.5 + Schroeder §7.1 — 2 hrs

Problems (a) derive $\langle N\rangle, \langle(\Delta N)^2\rangle$ for ideal gas from Ξ; (b) Langmuir adsorption (surface sites); (c) verify $\mu = (\partial F/\partial N)_{T,V}$ canonical vs. grand (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 5. ⋆ Bose–Einstein & Fermi–Dirac

Reading Schroeder §7.2–7.4 + Pathria Ch. 6 §6.1–6.3 + Ch. 7 §7.1 + Ch. 8 §8.1 — 2 hrs

Problems (a) degenerate Fermi gas: electrons in a metal, find E_F and $P(T{=}0)$; (b) photon gas: derive Stefan–Boltzmann from Ξ (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 6. ⋆ Phase transitions + Landau MF

Reading Pathria Ch. 11 §11.1–11.5 + Ch. 12 §12.1–12.2 + Schroeder §5.3 — 2 hrs

Problems (a) mean-field T_c for Ising + magnetisation $m(T)$ near T_c; (b) critical exponents $\alpha, \beta, \gamma, \delta$ in Landau expansion; (c) Clausius–Clapeyron coexistence (vapour pressure) (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 7. ⋆ Brownian motion + random walks

Reading Pathria Ch. 15 §15.1–15.4 — 2 hrs

Problems (a) Langevin equation: $\langle x^2(t)\rangle$ ballistic → diffusive crossover; (b) 1D random walk: diffusion coefficient + mean first-passage time (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 8. Synthesis cheat sheet

Reading Re-read summary sections of each chapter — 2 hrs

Problems Write 1-page SM cheat sheet: ensembles, free energies, derivatives, fluctuation–dissipation (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 9. Buffer week (catch-up + bonus)

Reading Revisit weakest of weeks 2–7 (2 hrs)

Problems One extra problem from each weak chapter (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Exp HEP — Part 1: Detectors, Statistics, Landscape

Wk 1. Interactions in matter (refresh)

Reading Grupen & Shwartz Chs. 1–2 + PDG "Passage of Particles Through Matter" — 2 hrs

Problems (a) PDG-based muon energy loss in 1 m of lead; (b) Cherenkov threshold for π/K in $n = 1.05$ aerogel (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 2. Trackers (gaseous & silicon)

Reading Grupen & Shwartz Chs. 3–4 + BRR excerpts on drift chambers — 2 hrs

Problems (a) design Si pixel layout for ATLAS ITk inner layer given radius / occupancy / fluence; (b) compute hit resolution for 50 μm pitch + binary readout (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 3. Calorimetry

Reading Grupen & Shwartz Chs. 7–8 + Wigmans RMP review — 2 hrs

Problems (a) sampling-cal energy resolution for 50 GeV e/π; (b) compute Molière radius / interaction length for lead vs. tungsten (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 4. Particle ID & particle flow

Reading Grupen & Shwartz Ch. 9 + CMS PF paper (JINST 2017) — 2 hrs

Problems Sketch the PF reconstruction chain end-to-end; identify the three dominant sources of confusion in jet PF (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 5. Trigger & DAQ

Reading ATLAS & CMS L1/HLT papers — 2 hrs

Problems Design a minimal trigger menu for $\mathcal{L} = 10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$: single e, μ, jet, E_T^{miss} thresholds + rate budget (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 6. Pileup, luminosity, backgrounds

Reading ATLAS / CMS pileup notes (post-LS2 docs) — 2 hrs

Problems Compute pileup at $\mu = 200$ for HL-LHC; consequence for jet-mass resolution and PUPPI mitigation (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 7. Statistics & HEP analysis

Reading Cowan Chs. 1–6 + PDG "Statistics" review — 2 hrs

Problems Profile-likelihood fit in Python: toy bg+signal model, extract signal strength $\hat{\mu}$ with one nuisance (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 8. Landscape map (2024–2035)

Reading Snowmass 2021 + P5 2023 + ESPP 2020 update — 2 hrs

Problems One-page table: Belle II, JUNO, HK, DUNE, LZ/XENONnT, LEGEND, EIC — timing + physics goal + my fit (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Wk 9. Positioning memo draft

Reading Skim 1–2 recent HL-LHC Phase-II TDR sections (preview for Summer #2) — 2 hrs

Problems Draft 2-page memo: *What my RHI background buys me for HEP* (hardware / software / stats) (1.3 hrs)

Time Reading 2.0 + Problems 1.3 = **3.3 hrs**

Summer #2

Jul 22 – Aug 18, 2026 · 4 weeks · 10 hrs/wk · ∼5 hrs/subj/wk

Nuc #1 — Nuclear Foundations

Anchor: re-cement decay modes, reaction kinematics, fission/fusion, detector mapping.

Wk 1. Beta + Gamma decay

Reading Loveland Chs. 8 (21 pp), 9 (27 pp) — 3 hrs

Problems Loveland 8.3, 8.5, 9.2; sketch β-spectrum from scratch + state multipole selection rules (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 2. Nuclear reactions

Reading Loveland Ch. 10 (full, ∼48 pp) — 3 hrs

Problems Loveland 10.3, 10.5; derive Rutherford $d\sigma/d\Omega$ from scratch; compute one Q-value (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 3. Fission & nuclear astrophysics

Reading Loveland Chs. 11 (30 pp) + 12 (32 pp) — 3 hrs

Problems Loveland 11.4 (asymmetric fragments); 12.3 (pp-chain); sketch binding-energy curve + mark fission, fusion, iron peak (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 4. Radiation detectors

Reading Loveland Ch. 18 (40 pp) — 3 hrs

Problems Loveland 18.2, 18.4; build a detector-to-particle / energy reference card (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Exp HEP — Part 2: Accelerators + TDR Deep-Dives

Wk 1. Accelerator I: beam dynamics

Reading Wiedemann Chs. 1–4 — 3 hrs

Problems (a) compute betatron tune for a FODO cell; (b) closed-orbit correction with one dipole kick (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 2. Accelerator II: RF, longitudinal, SR, luminosity

Reading Wiedemann Chs. 5–9 (selective) — 3 hrs

Problems (a) synchrotron equation: phase-stability analysis at LHC parameters; (b) compute HL-LHC luminosity from β^*, ϵ_n, f, N_b (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 3. TDR deep-dive #1

Reading Pick one: ATLAS ITk *or* CMS HGCAL *or* sPHENIX TDR — 3 hrs

Problems Write 5-pg TDR summary: design + 3 tradeoffs + one open question (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Wk 4. ⋆ TDR deep-dive #2 + capstone

Reading Second TDR (different sub-system) + FCC-ee CDR / muon collider docs (skim) — 3 hrs

Problems 6–8 pg capstone: critical TDR review + one area where you could contribute (2 hrs)

Time Reading 3.0 + Problems 2.0 = **5 hrs**

Combined milestones: profile-likelihood fit fluent; HI vs. pp detector differences articulated with numbers; two TDRs internalized.

Fall #1

Sep – Oct 2026 · 9 weeks · 20 hrs/wk · ∼6.7 hrs/subj/wk

Quantum Mechanics — Merzbacher Review

Anchor: 1D, central, spin to qual-speed; Pauli algebra reflexive by Wk 8.

Wk 1. Wave mechanics + Schrödinger

Reading Merzbacher Chs. 1–3 — 4 hrs

Problems (a) Gaussian wave packet: compute $\langle x\rangle, \Delta x, \Delta p$ and time evolution; (b) commutator algebra from Ch. 3 problems (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 2. Operators + SHO + gauge

Reading Merzbacher Chs. 4 §4.1–4.7, 5 — 4 hrs

Problems (a) SHO ladder: compute $\langle n|x^2|n\rangle$ and matrix element $\langle n|x|n{+}1\rangle$; (b) gauge transformation problem from Ch. 4 (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 3. 1D constant potentials + WKB

Reading Merzbacher Chs. 6 + 7 — 4 hrs

Problems (a) T, R for square barrier (find graphically + numerically); (b) WKB bound states for $V = \frac{1}{2}m\omega^2x^2 + \lambda x^4$; (c) α-decay tunneling estimate (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 4. Variational + perturbation; Hilbert formalism

Reading Merzbacher Chs. 8 §8.1–8.4 + 9 + 10 §10.1–10.5 — 4 hrs

Problems (a) variational ground-state of He with simple trial ψ; (b) first-order PT problem from Ch. 8; (c) position $\leftrightarrow$ momentum basis change (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 5. Angular momentum

Reading Merzbacher Ch. 10 §10.6–7 + Ch. 11 §11.1–11.4 — 4 hrs

Problems (a) raising / lowering between $|\ell, m\rangle$ states; (b) compute $\langle Y_{\ell m}|\cos\theta|Y_{\ell' m'}\rangle$; (c) coherent-state problem (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 6. ⋆ Central potentials & hydrogen

Reading Merzbacher Ch. 12 (full, read twice) — 4 hrs

Problems (a) 3D spherical well: find bound states for given depth and radius; (b) hydrogen: compute $\langle r\rangle, \langle 1/r\rangle, \langle 1/r^2\rangle$ in $n = 1, 2$; (c) 3D isotropic SHO: Cartesian + radial, confirm degeneracies (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 7. Scattering

Reading Merzbacher Ch. 13 §13.1–13.5 — 4 hrs

Problems (a) Born approximation for Yukawa / screened Coulomb; (b) hard-sphere low-energy phase shift; (c) verify optical theorem (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 8. ⋆ Spin

Reading Merzbacher Ch. 16 §16.1–16.6 — 4 hrs

Problems (a) prove $\{\sigma_i, \sigma_j\} = 2\delta_{ij}I$ and $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$, derive $\exp(i\theta\,\sigma\cdot\hat{n}/2)$; (b) spin-1/2 in $\vec{B}(t)$: Larmor precession + Rabi oscillations; (c) Bloch-sphere rotation sequence (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 9. Symmetries + time-dep PT + synthesis

Reading Merzbacher Chs. 17, 18, 19 (§17.1–5, §18.1–4, §19.1–5) — 4 hrs

Problems (a) Clebsch–Gordan: combine j_1=1 and j_2=1/2 explicitly; (b) Stark effect in hydrogen (2nd order); (c) Fermi's golden rule under harmonic perturbation; close with 1-page QM cheat sheet (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

RHI & Jets

Anchor: working Pythia 8 + FastJet pipeline; substructure understood end-to-end; jet quenching at parametric level.

Wk 1. Collider QCD foundations

Reading ESW Chs. 1–3 + PDG QCD + Salam TASI 2010 §1 — 4 hrs

Problems (a) derive factorization theorem schematically; (b) run $\alpha_s(Q^2)$ at 1-loop and plot from 1 GeV to 1 TeV (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 2. PDFs & DGLAP

Reading ESW Ch. 4 + LHAPDF docs (CT18NNLO) — 4 hrs

Problems (a) plot CT18 PDFs at $Q = 2, 10, 100, 1000$ GeV on one canvas; (b) ESW DGLAP splitting-function problem (one) (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 3. Jet algorithms + Pythia/FastJet setup

Reading MSS Ch. 2 + FastJet manual §1–3 — 4 hrs

Problems (a) implement anti-k_T in pure Python for ∼50 particles; (b) compare to `fastjet` package on same input; (c) install Pythia 8 + LHAPDF + uproot pipeline (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 4. Pythia 8 showers; NLO intro

Reading Pythia worksheet + Sjöstrand reviews + ESW Ch. 5 — 4 hrs

Problems Run Pythia pp→dijet at $\sqrt{s} = 13$ TeV; plot inclusive jet p_T; overlay public ATLAS/CMS data (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 5. Substructure I: grooming

Reading MSS Chs. 3–4 + Soft Drop paper (Larkoski et al. 2014) — 4 hrs

Problems Generate W→ qq' and QCD dijet samples; apply mMDT + Soft Drop; plot groomed mass; quantify W/QCD separation (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 6. Substructure II: N-subj, ECFs, tagging

Reading MSS Chs. 5–7 + Thaler–Van Tilburg — 4 hrs

Problems Compute τ_{21} for W and QCD samples from Wk 5; build simple boosted-W tagger + plot ROC curve (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 7. QGP foundations

Reading Busza–Rajagopal–van der Schee ARNPS 2018 + Florkowski Chs. 1–4 — 4 hrs

Problems (a) derive Stefan–Boltzmann energy density for $n_f = 3$ QGP; (b) viscous $T^{\mu\nu}$ derivation and write $\partial_\mu T^{\mu\nu} = 0$ (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 8. ⋆ Jet quenching + observables

Reading Mehtar-Tani–Milhano–Tywoniuk Jet Quenching Primer + ATLAS/CMS PbPb jet papers — 4 hrs

Problems (a) numerical $\langle \Delta E \rangle$ for fixed $\hat{q}$, L; (b) reproduce jet-R_{AA} vs. p_T qualitatively from a simple energy-loss model (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 9. Synthesis; portfolio piece

Reading MSS Ch. 8 + Connors–Nattrass–Reed–Salur RMP 2018 — 4 hrs

Problems Draft 4–6 pg note: *A First Pass at Jet Substructure, from pp to PbPb* (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

E&M — Zangwill Review

Wk 1. Math + Maxwell intro

Reading Zangwill Chs. 1 §1.1–1.6 + 2 — 4 hrs

Problems (a) one δ-function / Green's theorem problem from Ch. 1; (b) Ch. 2 unit-conversion or Maxwell derivation problem (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 2. Electrostatics + multipoles

Reading Zangwill Chs. 3 §3.1–3.7 + 4 §4.1–4.6 — 4 hrs

Problems (a) Gauss-law for non-uniform cylindrical charge distribution; (b) compute Cartesian + spherical multipole moments of a given distribution (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 3. Laplace + BVPs + images / Green fn

Reading Zangwill Chs. 7 §7.1–7.7 + 8 §8.1–8.5 — 4 hrs

Problems (a) grounded sphere in uniform external field (separation of variables); (b) point charge near a grounded plane (image method); (c) construct Dirichlet Green function for half-space (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 4. Magnetostatics + magnetic multipoles + forces

Reading Zangwill Chs. 10 §10.1–10.5 + 11 §11.1–11.3 + 12 §12.1–12.6 — 4 hrs

Problems (a) toroid with varying current density via Biot–Savart; (b) cyclotron motion in $\vec{B}$; (c) force between two parallel current loops (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 5. Dynamic & quasistatic fields

Reading Zangwill Ch. 14 §14.1–14.10 — 4 hrs

Problems (a) EMF in moving loop with changing flux (Faraday); (b) skin-depth: derive $\delta = \sqrt{2/\mu\sigma\omega}$ and compute for Cu at 1 GHz (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 6. Conservation laws + waves in vacuum

Reading Zangwill Chs. 15 §15.1–15.5 + 16 §16.1–16.4 — 4 hrs

Problems (a) Poynting-vector energy-conservation for a coaxial cable; (b) polarization: convert circular $\leftrightarrow$ linear; compute Stokes parameters (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 7. Waves in matter; Fresnel

Reading Zangwill Ch. 17 §17.1–17.4 — 4 hrs

Problems (a) Fresnel R, T for s- and p-polarization at oblique incidence; (b) verify Brewster's angle from $r_p = 0$; (c) total internal reflection threshold problem (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 8. Retarded potentials + radiation

Reading Zangwill Chs. 20 §20.1–20.5 + 21 §21.1–21.4 — 4 hrs

Problems (a) Larmor formula for accelerating charge in circular motion; (b) Thomson cross-section derivation; (c) Rayleigh ω^4 scattering — show why sky is blue (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Wk 9. $\star$ Special relativity in E&M

Reading Zangwill Ch. 22 §22.1–22.7 — 4 hrs

Problems (a) Lorentz-transform $\vec{E}$ and $\vec{B}$ between frames; (b) verify $F^{\mu\nu}$ form of Maxwell's equations in 5 minutes (2.7 hrs)

Time Reading 4.0 + Problems 2.7 = **6.7 hrs**

Fall #2

Nov – Dec 2026 · 9 weeks · 20 hrs/wk · QFT #1 (12 hr/wk) + Classical Pt 1 (8 hr/wk)

QFT I is the densest class in the plan. Aim 7 hrs reading + 5 hrs problems / week. Classical runs in the background at 5 + 3 hrs.

QFT #1 — Canonical Formalism through Tree-Level QED

Primary spine: Peskin & Schroeder Chs. 1–6. Schwartz as second voice; Tong notes as on-ramp.

Wk 1. Lorentz group + classical field theory; scalar quantization

Reading P&S §2.1–2.4; Tong Ch. 1 + §2.7–2.9 — 7 hrs

Problems P&S 2.1 (classical EM as field theory); 2.2 (complex scalar charge current); derive KG from action; mode expansion (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 2. Propagators & causality

Reading P&S §2.4 (revisit); Tong §2.7–2.9 (full) — 7 hrs

Problems Derive Feynman propagator from contour integration; verify causality for spacelike separation; one Schwartz Ch. 3 problem (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 3. Weyl & Dirac spinors; discrete symmetries

Reading P&S §3.1–3.6 + Schwartz Ch. 11 — 7 hrs

Problems P&S 3.1 (Lorentz invariance of Dirac $\mathcal{L}$); 3.4 (Majorana fermions); verify C, P, T on bilinears explicitly (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 4. Dirac field quantization

Reading P&S §3.5 + Schwartz Ch. 12 — 7 hrs

Problems Derive canonical anticommutators; spin–statistics from positivity of H; one Schwartz Ch. 12 problem (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 5. Interacting fields; Dyson + Wick + Feynman diagrams

Reading P&S §4.1–4.4 — 7 hrs

Problems P&S 4.1 (source-created field); 4.3 (linear sigma model); derive S-matrix from Dyson; draw all ϕ^4 diagrams to 2nd order (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 6. QED Feynman rules + $\mathbf{e^+e^- \to \mu^+\mu^-}$

Reading P&S §4.8 + 5.1 — 7 hrs

Problems P&S 5.1 (Coulomb scattering); derive QED Feynman rules from $\mathcal{L}_{\text{QED}}$ on a blank page; compute $d\sigma/d\Omega$ for $e^+e^- \to \mu^+\mu^-$ end-to-end (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 7. Compton + crossing symmetry

Reading P&S §5.5 — 7 hrs

Problems P&S 5.3 (positronium lifetimes); derive Klein–Nishina cross section; demonstrate crossing symmetry for Compton vs. e^+e^- pair production (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 8. Bhabha, Møller, LSZ

Reading P&S §4.6 + Schwartz Ch. 6 — 7 hrs

Problems LSZ reduction derivation; compute Bhabha + Møller amplitudes; one Schwartz Ch. 10 / 13 problem (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 9. ⋆ Capstone — $e^+e^- \to q\bar{q}$ tree level

Reading Re-read P&S Ch. 5; PDG LEP R-ratio review — 7 hrs

Problems Write 6–8 pg LaTeX: extract Feynman rules; compute amplitude with trace technology; derive $d\sigma/d\Omega$; include N_c; plot $\sigma(\sqrt{s})$ for u, d, c, b; compare LEP R-ratio data; discuss threshold + QCD corrections (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Classical Pt 1 — Lagrangian → Rigid Body

Anchor: write $\mathcal{L} = T - V$ in arbitrary coordinates without thinking; full comfort with rotations and rigid-body dynamics. Lemos primary, Taylor for intuition, Goldstein Ch. 4 for rotation tensors.

Wk 1. Lagrangian basics (1)

Reading Lemos §1.1–1.3 (Newtonian review, constraints, d'Alembert) — 5 hrs

Problems (a) bead on a rotating wire (constraints + generalized coords); (b) Lemos 1.3 (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 2. Lagrangian basics (2)

Reading Lemos §1.4–1.5 + Taylor Ch. 7 (for intuition) — 5 hrs

Problems (a) Lemos 1.6 (Atwood machine variant); (b) Lemos 1.9 or 1.11 (double pendulum) (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 3. Central forces

Reading Lemos §1.6–1.7 (generalized potentials, Bertrand) — 5 hrs

Problems (a) Lemos 1.14 (Kepler orbit); (b) one turning-point problem from Ch. 1 (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 4. Variational principle + Noether

Reading Lemos Ch. 2 (full: variational, non-holonomic, Noether) — 5 hrs

Problems (a) Lemos 2.5 (Noether-style problem); (b) Lemos 2.8; (c) derive conserved quantity from a symmetry (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 5. ⋆ Rotations (1)

Reading Lemos §3.1–3.3 (orthogonal transformations, Euler angles) + Goldstein Ch. 4 — 5 hrs

Problems (a) Euler-angle transformation: rotate a vector via two composed rotations; (b) Lemos Ch. 3 problem on infinitesimal generators (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 6. ⋆ Rotations (2) — non-inertial frames

Reading Lemos §3.4–3.6 (rotating frames, Coriolis, centrifugal) — 5 hrs

Problems (a) Foucault pendulum; (b) Coriolis deflection of a projectile at given latitude; (c) prove $[L_i, L_j]_{\text{PB}} = \varepsilon_{ijk} L_k$ classically (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 7. Rigid-body dynamics (1)

Reading Lemos Ch. 4 §4.1–4.5 (inertia tensor, principal axes) — 5 hrs

Problems (a) compute inertia tensor for cube about a face vs. vertex; (b) diagonalize a non-trivial I_{ij} (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 8. Rigid-body dynamics (2)

Reading Lemos §4.6–4.9 (Euler's equations, symmetric top) — 5 hrs

Problems (a) symmetric top: derive precession + nutation; (b) torque-free symmetric body free-rotation problem (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 9. Review + synthesis

Reading Re-read worked examples from Chs. 1–4 — 5 hrs

Problems Bring forward 3 problems from prior weeks that gave trouble; redo cold (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Spring #1 *Jan – Feb 2027 · 9 weeks · 20 hrs/wk · QFT #2 (12 hr/wk) + Classical Pt 2 (8 hr/wk)*

> QFT II Weeks 3–5 (renormalization) are the crunch. Pull from CM slack week if needed — never the reverse.

QFT #2 — Path Integrals, Renormalization, Standard Model

Primary: P&S Chs. 9–11, 15–17, 20. Schwartz preferred for renormalization.

Wk 1. Path integrals (scalars, fermions, gauge)
Reading P&S §9.1–9.5 — 7 hrs
Problems P&S 9.1 (retarded prop from PI); QED PI with gauge fixing; Grassmann Gaussian (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 2. One-loop in ϕ^4; regularization
Reading P&S §10.1–10.2 — 7 hrs
Problems P&S 10.2 (alternatives to dim reg); compute ϕ^4 at 1-loop with cutoff (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 3. ⋆ QED 1-loop: vacuum polarization, self-energy, vertex
Reading P&S §10.3 + Schwartz Ch. 16 — 7 hrs
Problems P&S 10.3 (mass renormalization); compute $\Pi(q^2)$ at 1-loop; derive $g-2$ at 1-loop (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 4. ⋆ RG; Callan–Symanzik
Reading P&S §12.1–12.2 + Schwartz Ch. 23 — 7 hrs
Problems P&S 12.2 (β in Yukawa theory); compute $\beta(e)$ at 1-loop; solve RG to get $\alpha(\mu)$ (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 5. Non-Abelian gauge theory; Faddeev–Popov, BRST
Reading P&S §15.1–15.3, 16.1–16.3 — 7 hrs
Problems P&S 16.1 (Arnowitt–Fickler gauge); derive covariant D_μ for SU(N); write QCD Feynman rules from scratch (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 6. Asymptotic freedom
Reading P&S §16.5, 16.7 — 7 hrs
Problems P&S 16.3 (1-loop gluon self-energy); compute $\beta(g_s)$ at 1-loop including ghosts; solve RG to plot $\alpha_s(\mu)$ from m_Z to 1 TeV vs. PDG (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 7. SSB; Goldstone; Higgs mechanism; electroweak
Reading P&S §11.1 + 20.1–20.2 — 7 hrs
Problems Linear sigma in Higgs phase; derive W, Z masses from $\mathcal{L}_{\text{EW}}$ via g, g', v (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 8. Flavor + CKM + triple gauge
Reading P&S §20.3 — 7 hrs
Problems P&S 20.2 (Higgs decays to fermions); P&S 20.3 (triple gauge couplings) (5 hrs)
Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 9. ⋆ Capstone — $\alpha_s(\mu)$ + Higgs ggF

Reading Re-read renormalization + EW; ATLAS or CMS H$\to\gamma\gamma$ paper — 7 hrs

Problems ∼10 pg writeup: derive QCD β including ghosts; plot $\alpha_s(\mu)$; derive Higgs ggF at LO; estimate at $\sqrt{s} = 13.6$ TeV vs. ATLAS/CMS; discuss H$\to\gamma\gamma$ or y_t measurement (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Classical Pt 2 — Small Oscillations → Hamiltonian

Wk 1. Small oscillations (1) — 1D + stationary motion

Reading Lemos §5.1–5.3 + Taylor Ch. 11 (intuition) — 5 hrs

Problems (a) 1D oscillation about equilibrium for given $V(x)$; (b) Lemos 5.x: stability analysis problem (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 2. ⋆ Small oscillations (2) — 2-DOF normal modes

Reading Lemos §5.4–5.5 — 5 hrs

Problems (a) double pendulum: find M and K matrices, diagonalize, normal frequencies; (b) coupled-mass-spring system, 2 modes (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 3. ⋆ Small oscillations (3) — general N-DOF

Reading Lemos §5.6 + Goldstein Ch. 6 (vibrations) — 5 hrs

Problems (a) triatomic molecule (linear, 3 atoms) — use symmetry to guess modes, then verify; (b) compute normal coordinates explicitly (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 4. Damped + driven oscillations

Reading Lemos extension + Taylor Ch. 5 — 5 hrs

Problems (a) damped-driven coupled system; resonance + Q-factor; (b) parametric oscillator stability problem (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 5. Hamiltonian formalism (1)

Reading Lemos Ch. 7 §7.1–7.3 (canonical eqns, symmetries, virial) — 5 hrs

Problems (a) write H for a charged particle in EM field; (b) virial theorem applied to oscillator + Kepler (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 6. Hamiltonian formalism (2) — generating functions

Reading Lemos Ch. 8 §8.1–8.3 — 5 hrs

Problems (a) verify $Q = \sqrt{2p}\cos q$, $P = \sqrt{2p}\sin q$ is canonical via generating function; (b) construct $F_2(q, P, t)$ for harmonic oscillator (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 7. Poisson brackets

Reading Lemos §8.4 — 5 hrs

Problems (a) compute $\{L_i, L_j\}_{\text{PB}}$ and verify angular momentum algebra; (b) time evolution of $\vec{L}\cdot\hat{n}$ for a particle in central potential (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 8. Infinitesimal canonical transformations + symmetries

Reading Lemos §8.5; skim §8.7–8.9 — 5 hrs

Problems (a) generator of rotations from Poisson brackets; (b) connect Noether (Wk 4 of Pt 1) to canonical generators of symmetries (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 9. Synthesis

Reading Re-read Chs. 5, 7, 8 examples — 5 hrs

Problems Write 1-page CM cheat sheet: $\mathcal{L} \to H$ Legendre transform, canonical eqns, PB identities, normal-mode recipe (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Spring #2

Mar – May 2027 · 13 weeks · 20 hrs/wk · Astro (12 hr/wk) + Optics (8 hr/wk)

> The goal here is literacy and motivation, not mastery. Aim to (a) follow an astro colloquium, (b) read SiPM / Cherenkov papers fluently, (c) identify one or two bridge topics to follow deeper.

Astro & Nuclear Astrophysics

Read Class-4 Foundations (Parts A–D) before Wk 1. Maoz primary; Phillips alongside; Iliadis for nuclear weeks; Perkins for particle astro; Ryden for cosmology.

Wk 1. Observational astronomy + HR + foundations

Reading Foundations Part A; Maoz Ch. 1; C&O Chs. 1–3 — 7 hrs

Problems (a) Maoz 1.x scales / distances problems (3 problems); (b) place Sun on HR diagram; estimate distance from parallax (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 2. Radiation + stellar parameters

Reading Foundations B.1; Maoz Ch. 2; Phillips Chs. 1–2 — 7 hrs

Problems (a) Stefan–Boltzmann: L from R and $T_{\rm eff}$ for 3 spectral types; (b) Wien displacement: peak wavelength for sun vs. 5000 K vs. 30000 K star (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 3. Stellar structure + virial

Reading Foundations B.2–B.4; Maoz Ch. 3; Phillips Chs. 3–5 — 7 hrs

Problems (a) Lane–Emden polytrope for $n = 3/2$ (white dwarf); (b) virial-theorem estimate of $T_{\rm interior}$ for Sun; (c) Phillips ch. 3 problem on hydrostatic equilibrium (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 4. Nuclear energy in stars (concept)

Reading Maoz Ch. 4; Phillips Chs. 6–7; Iliadis Ch. 1 — 7 hrs

Problems (a) pp-chain rate estimate at $T = 1.5 \times 10^7$ K; (b) compute Q-value for ${}^1\mathrm{H} + {}^1\mathrm{H} \to {}^2\mathrm{H} + \mathrm{e}^+ + \nu$ (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 5. Nuclear reactions (depth)

Reading Iliadis Chs. 2–4 + Rolfs–Rodney supplement — 7 hrs

Problems (a) draw CNO cycle flow diagram with rates; (b) explain why CNO dominates in massive stars; (c) Iliadis astro-numerics problem (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 6. Stellar evolution + nucleosynthesis (s, r)

Reading C&O Chs. 13–14; Iliadis Ch. 5 — 7 hrs

Problems (a) sketch s-process path on chart of nuclides; (b) identify r-process site candidates and their signatures (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 7. Supernovae + compact-object formation

Reading C&O Ch. 15; Bethe reviews; Maoz Ch. 5 — 7 hrs

Problems (a) ν-driven mechanism summary: 5-step writeup; (b) estimate ν flux on Earth from a Galactic core-collapse SN at 10 kpc vs. Super-K threshold (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 8. ⋆ Compact objects: WDs, NSs, BHs

Reading Foundations B.5–B.6; Maoz Ch. 4; Shapiro–Teukolsky Chs. 2–3, 8–9 — 7 hrs

Problems (a) derive Chandrasekhar mass and discuss composition independence; (b) write TOV equation; (c) compute Eddington luminosity for $1M_\odot, 10^6 M_\odot$ accretors (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 9. Mergers + multimessenger

Reading S–T Ch. 12; GW170817 multimessenger paper; Kasen kilonova — 7 hrs

Problems (a) explain kilonova / r-process link (one page); (b) connect GW chirp mass to NS–NS vs. NS–BH (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 10. Cosmic rays + ISM + astrophysical neutrinos

Reading Perkins Chs. 6–9; Gaisser excerpts; IceCube diffuse-flux papers; Maoz Chs. 6, 10 — 7 hrs

Problems (a) plot CR spectrum on log–log; annotate knee, ankle, GZK; (b) sketch IceCube energy-sensitivity curve and explain ν flavor mix (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 11. ⋆ Cosmology (Friedmann, BBN, CMB)

Reading Foundations B.7; Ryden Chs. 1–5; Dodelson–Schmidt Chs. 1–2; Planck 2018 — 7 hrs

Problems (a) Ryden end-of-chapter problem on Friedmann + critical density; (b) BBN ^{4}He / H freeze-out from Boltzmann equation; (c) sketch CMB power spectrum + identify acoustic peak (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 12. Dark matter / dark energy

Reading Perkins; Kolb–Turner relevant chapters; XENONnT, LZ papers — 7 hrs

Problems (a) derive WIMP thermal relic abundance (Kolb–Turner); (b) read one DM direct-detection paper, summarize sensitivity vs. neutrino floor (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Wk 13. ⋆ Capstone — RHI/HEP × astro bridge

Reading Pick bridge topic and review — 7 hrs

Problems Write 6-page review on chosen bridge: kilonova r-process; NS EOS from lattice QCD; sterile ν in cosmology; axion DM searches; MM constraints on NS merger remnants (5 hrs)

Time Reading 7.0 + Problems 5.0 = **12 hrs**

Optics — Classical, Fourier, Quantum

Wk 1. Geometrical optics + matrix methods

Reading Hecht Chs. 5–6 — 5 hrs

Problems (a) ray-transfer matrices: trace ray through compound lens system; (b) design a simple 2-lens imaging system with given magnification (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 2. Wave optics + interference

Reading Hecht Chs. 2–3 + Ch. 9 — 5 hrs

Problems (a) Michelson interferometer: visibility + fringe spacing vs. source bandwidth; (b) Fabry–Perot finesse for given R (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 3. Diffraction I (Fraunhofer)

Reading Hecht Ch. 10 + Goodman Ch. 4 — 5 hrs

Problems (a) compute Fraunhofer pattern of rectangular and circular aperture; (b) compare to numerical FFT (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 4. Diffraction II (Fresnel) + Fourier optics

Reading Goodman Chs. 4–6 — 5 hrs

Problems (a) Implement Fresnel pattern in Python (beam → aperture → lens → image); (b) explain spatial filtering on a 4-f system (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 5. Polarization (Jones, Stokes)

Reading Hecht Ch. 8 + Saleh–Teich Ch. 6 — 5 hrs

Problems (a) write Jones matrices for QWP, HWP, linear polarizer at angle θ; (b) compute Stokes parameters for given $\vec{E}$; (c) verify Mueller matrix for one optical element (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 6. Gaussian beams

Reading Saleh–Teich Ch. 3 — 5 hrs

Problems (a) derive q-parameter; (b) apply ABCD law for beam through thick lens; (c) compute Rayleigh range for a focused 1 mW HeNe beam (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 7. Laser resonators & modes

Reading Saleh–Teich Chs. 10–11 — 5 hrs

Problems (a) design a stable confocal resonator; (b) compute longitudinal and transverse mode spacing (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 8. Laser physics + amplification

Reading Saleh–Teich Chs. 13–14 + Siegman excerpts — 5 hrs

Problems (a) rate-equation model of He–Ne laser; population inversion threshold; (b) saturation intensity for a 4-level laser (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 9. ⋆ Quantum optics I: photons, coherent states

Reading Loudon Chs. 4–5 + Saleh–Teich Ch. 12 — 5 hrs

Problems (a) compute Poissonian photon statistics for a coherent state; (b) show $\langle\alpha|\alpha\rangle = 1$ and $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 10. ⋆ Quantum optics II: coherence & detection

Reading Loudon Chs. 3, 6 + Gundacker–Heering SiPM RMP — 5 hrs

Problems (a) compute $g^{(2)}(\tau)$ for coherent state and for thermal state; (b) explain HBT intensity-correlation result (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 11. HEP-relevant detector optics

Reading SiPM / Cherenkov-imaging papers (Gundacker–Heering RMP 2020 + RICH chapter) — 5 hrs

Problems (a) summarize how SiPM dark count rate scales with bias and temperature; (b) compute Cherenkov ring radius for $\beta = 0.99$ in $n = 1.05$ aerogel (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 12. ⋆ Capstone — HEP-relevant optics review

Reading Pick topic; gather references — 5 hrs

Problems 5–6 pg review on chosen topic: SiPMs in HEP calorimetry, Cherenkov imaging at IceCube/KM3NeT, atomic clocks for dark-matter searches, or Fabry–Perot cavities in GW interferometers (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Wk 13. Review + synthesis

Reading Re-read class notes; skim Loudon + Saleh–Teich highlights — 5 hrs

Problems Write 1-page Optics cheat sheet across topics (ray, Fourier, Gaussian, Jones, $g^{(2)}$) (3 hrs)

Time Reading 5.0 + Problems 3.0 = **8 hrs**

Closing Notes

Slip Plan (priority order)

- **Drop reading before problems.** Retrieval practice builds recall; reading alone fades.
- **Protect one continuous hour per subject per week.** Three 20-min sessions do not equal one 60-min session.
- **Every 4 weeks, re-scan margin notes.** Anything you cannot recall within 30 seconds goes on a "revisit" list.
- **Never drop:** Higgs (Mann 23), neutrino oscillations (Thomson 13), renormalization (P&S 10, 12), spin (Merzbacher 16), compact objects (Maoz 4 / S–T 8–9), normal modes (Lemos 5), Fresnel + retardation (Zangwill 17, 20).

If you finish early in a period

- Go deeper into the chosen topic. Revisit cut chapters (aberrations, waveguides, Hamilton–Jacobi).
- Do not add new books — going deep beats going wide.
- Write a 1-page synthesis tying the period together.

Hour accounting — summary

Period	Wks × hrs/wk	Total period	Per subject (3 or 2 subj)
Summer #1	9×10	90 hrs	≈30 hrs each (3 subj)
Summer #2	4×10	40 hrs	≈20 hrs each (2 subj)
Fall #1	9×20	180 hrs	≈60 hrs each (3 subj)
Fall #2	9×20	180 hrs	QFT 108 / CM 72
Spring #1	9×20	180 hrs	QFT 108 / CM 72
Spring #2	13×20	260 hrs	Astro 156 / Optics 104
Total	**53 weeks**	**∼930 hrs**	—

All weekly entries above sum to the listed per-subject weekly slice. Reading-to-problem ratio is held at ≈60/40 throughout. If a week's actual time differs from plan, the slip rules above apply.